

Platform Membership, Marketing-Mix Effectiveness, and Capacity Utilisation

Model Tests, Extended Features, and Interpretation

JAMS Platform Membership Analysis

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1 Purpose

This document summarizes the current empirical evidence for the platform membership paper. The objective is not to test whether platform membership mechanically raises capacity utilisation. Instead, the analysis tests whether platform membership changes the effectiveness of marketing-mix instruments, especially pricing, promotion, place, product, and process-related features. The central dependent variable is transformed capacity utilisation, `ur_boxcox`. Trip volume, `log_trip_count_day`, is used as a secondary outcome for selected robustness checks.

The empirical panel is a city–operator–day panel for 2024, covering March 11, 2024 through August 22, 2024. The main full-sample models use 14,514 observations, 97 city–operator units, and 30 cities. Platform membership is coded as substantive bookable/payable integration and separated from weak exposure. Platform types distinguish large aggregator membership, local MaaS membership, and multi-platform membership.

2 Model Families

The empirical strategy uses a family of fixed-effect regressions with cluster-robust standard errors at the city–operator level. The two main fixed-effect strategies are:

- **Market FE:** city, date, and operator fixed effects.
- **Unit FE:** city–operator and date fixed effects.

The market-FE models identify differences across operators and cities after controlling for date shocks and operator baselines. The unit-FE models are stricter: they identify within-city–operator changes over time net of date fixed effects. Robust findings are therefore those that survive both market-FE and unit-FE specifications or are repeatedly confirmed across threshold, city-drop, outcome, and feature-extension tests.

The core model classes are:

1. Platform main-effect models:

$$Y_{ict} = \beta_1 \text{Platform}_{ict} + X_{ict}\Gamma + \alpha + \epsilon_{ict}.$$

2. Marketing-mix moderation models:

$$Y_{ict} = \beta_1 M_{ict} + \beta_2 Platform_{ict} + \beta_3 M_{ict} \times Platform_{ict} + X_{ict}\Gamma + \alpha + \epsilon_{ict}.$$

3. Platform-type threshold models:

$$Y_{ict} = \sum_k \beta_k HighUnlockFee_{ict} \times PlatformType_{ict}^k + X_{ict}\Gamma + \alpha + \epsilon_{ict}.$$

4. Extended feature models:

$$Y_{ict} = Feature_{ict} \times PlatformType_{ict} + X_{ict}\Gamma + \alpha + \epsilon_{ict}.$$

Controls include own price per minute, unlock fee, relative fleet size, coverage, exclusive coverage, promotion activity, and weather controls, depending on the model family. Extended feature tests additionally use same-mode competition, same-platform competition, active-vehicle shares, relative price gaps, commute/night usage shares, and externally coded product/process/pass features.

3 Feature Construction

The extended feature layer adds three kinds of variables.

Competition and market structure. Same-city and same-mode competitor counts measure direct competitive pressure. Concentration is captured through trip-share and active-vehicle-share Herfindahl-type indices. Same-mode measures are preferred because scooter-to-scooter competition is more meaningful than a pooled count of all shared mobility providers.

Relative price position. The most important new features compare each provider’s price to the cheapest same-mode competitor in the same city on the same day. The key variable is `unlock_fee_gap_to_same_mode_min_z`. It measures whether the provider has a salient upfront access-price disadvantage relative to direct same-mode alternatives. A parallel variable is constructed for price per minute.

Product, process, and pass affordances. The panel already contains group ride, vehicle-type breadth, subscription options, and complementary services. Additional provider-level external coding captures reservation availability, pass/free-unlock affordances, minute bundles, and mandatory parking/geofence rules. These external flags are useful for theory development, but because they vary mostly at the provider level, they are not yet cleanly estimable as causal city-date effects.

4 Results

Table 1: Relevant Findings, Robustness, and Full-Sample Estimates

Finding	Substantive claim	Robustness	Market FE	Unit FE	Observation basis	Primary source
F1	Platform membership does not create a stable generic utilisa- tion lift.	Not robust as main effect	0.061; $p = 0.008$	-0.010; $p = 0.205$	Full sample, $N = 14,514$. Positive only in market FE; unit FE not significant.	idea_tests/05
F2	Unlock fees become more detrimental in platform contexts.	Robust in market FE; weaker in strict unit FE	-0.0715; $p < 0.001$	-0.0236; $p = 0.349$	Full sample, $N = 14,514$; negative in 9 significant concept-family models.	concept_aligned
F3	High unlock fees are harmful in platform contexts.	Robust	-0.104; $p < 0.001$	-0.040; $p < 0.001$	Full sample, $N = 14,514$; negative in 6/6 threshold models.	deep_dive
F3/F4	High unlock fee $\times$ Local MaaS.	Very robust	-0.150; $p = 0.002$	-0.046; $p < 0.001$	Full sample, $N = 14,514$; negative in 6/6 models.	deep_dive
F3/F4	High unlock fee $\times$ Multi-platform.	Very robust	-0.131; $p = 0.007$	-0.098; $p < 0.001$	Full sample, $N = 14,514$; negative in 6/6 models.	deep_dive
F3	EUR 1.00 unlock-fee threshold $\times$ Local MaaS.	Very robust	-0.229; $p < 0.001$	-0.049; $p = 0.002$	Full sample, $N = 14,514$; Local MaaS high-fee support: 90 rows.	threshold
F3	EUR 1.00 unlock-fee threshold $\times$ Multi-platform.	Very robust	-0.146; $p < 0.001$	-0.099; $p < 0.001$	Full sample, $N = 14,514$; multi-platform high-fee support: 1,308 rows.	threshold
F3	EUR 1.20 unlock-fee threshold $\times$ Local MaaS.	Very robust	-0.233; $p < 0.001$	-0.047; $p = 0.006$	Full sample, $N = 14,514$; Local MaaS high-fee support: 72 rows.	threshold
F3	EUR 1.20 unlock-fee threshold $\times$ Multi-platform.	Very robust	-0.171; $p < 0.001$	-0.082; $p < 0.001$	Full sample, $N = 14,514$; multi-platform high-fee support: 1,029 rows.	threshold
F3a	Local MaaS high-fee penalty is not driven by one platform city.	Robust	Median city-drop coefficient at EUR 1.00: -0.209	Median city-drop coefficient at EUR 1.00: -0.035	13/13 leave-one-platform-city models negative and significant for EUR 0.99, 1.00, and 1.20.	city_sensitivity
F3a	Multi-platform high-fee penalty is not a single-city artifact.	Robust at EUR 1.00 and 1.20	Median city-drop coefficients: -0.107 at EUR 1.00; -0.135 at EUR 1.20	Median city-drop coefficients: -0.085 at EUR 1.00; -0.073 at EUR 1.20	13/13 leave-one-platform-city models negative and significant at EUR 1.00 and 1.20.	city_sensitivity

Finding	Substantive claim	Robustness	Market FE	Unit FE	Observation basis	Primary source
F4a	Large aggregator-only / Free Negative Now-style membership does not show a stable high-fee penalty.	evince	-0.045; $p = 0.066$	-0.016; $p = 0.325$	Full sample, $N = 14,514$; not robustly significant.	threshold
F3b	Local MaaS high-fee result gen- eralizes from utilisation to trip volume.	Robust for Local MaaS	Log trips, EUR 1.00: -1.014; $p < 0.001$	Log trips, EUR 1.00: -0.130; $p < 0.001$	Full sample, $N = 14,514$; not a pure utilisation-ratio arti- fact.	outcome
F3b	Multi-platform trip-volume evi- dence is weaker than utilisation evidence.	Moderate	Log trips, EUR 1.00: -0.538; $p < 0.001$	Log trips, EUR 1.00: -0.010; $p = 0.873$	Full sample, $N = 14,514$; strong in market FE, not in unit FE.	outcome
F5	Generic promotion $\times$ platform membership is positive mainly within unit.	Moderate	0.012; $p = 0.424$	0.0115; $p < 0.001$	Full sample, $N = 14,514$; pro- motion rows: 738.	promotion
F6	Subscriber-free-minutes promo- tions fit multi-platform expo- sure.	Exploratory mechanism, locally bounded	0.122; $p < 0.001$	0.043; $p < 0.001$	Full sample model, $N = 14,514$, but treated support is only Nürnberg-VOI, 144 rows.	deep_dive
F16	Vehicle-type breadth buffers Lo- cal MaaS high-fee penalties.	Moderate to ro- bust	0.074; $p < 0.001$	0.014; $p < 0.001$	Full sample, $N = 14,514$; Lo- cal MaaS high-fee support: 90 rows.	feature_fee
F19	Relative same-mode unlock-fee disadvantage $\times$ Local MaaS.	Very robust; strongest new feature result	-0.089; $p < 0.001$	-0.030; $p = 0.002$	Full sample, $N = 14,514$; fea- ture fully observed.	extended
F19	Relative same-mode unlock-fee disadvantage $\times$ Multi-platform.	Very robust	-0.091; $p < 0.001$	-0.041; $p < 0.001$	Full sample, $N = 14,514$; fea- ture fully observed.	extended
F20	Relative per-minute price disad- vantage is weaker than relative unlock-fee disadvantage.	Moderate	-0.104; $p = 0.003$	-0.020; $p = 0.153$	Full sample, $N = 14,514$; extended weaker and less robust than unlock-fee gap.	extended
F21	Same-mode competition is a Lo- cal MaaS boundary condition, not the main mechanism.	Moderate	-0.059; $p = 0.013$	-0.019; $p = 0.083$	Full sample, $N = 14,514$; extended same-mode competitor fea- ture fully observed.	extended
F22	Own same-mode active-vehicle share helps platform contexts but does not offset high fees.	Moderate	For multi-platform: -0.007; $p = 0.819$	For multi-platform: 0.019; $p = 0.024$	Full sample, $N = 14,514$; pos- itive mainly within unit.	extended
F23	Local MaaS performs worse in night-oriented usage contexts.	Moderate	-0.019; $p = 0.001$	-0.018; $p < 0.001$	Full sample, $N = 14,514$; extended night-trip share from raw trips.	extended

Finding	Substantive claim	Robustness	Market FE	Unit FE	Observation basis	Primary source
F24	Provider-level pass, free-unlock, reservation, and parking/geofence flags are useful but not robust estimators.	Identification caveat	Not interpreted due to rank/collinearity issues	Not interpreted due to weak within-unit variation	Full sample fields available, but scooter-only pass/minute-bundle support has only one unique value.	extended
F25	Same-platform competitor count is not estimable cleanly.	Not supported	Implausibly large coefficients missing p-values	Unstable and collinear	and Feature is mechanically tied to platform/city/operator structure.	extended

5 Neutral Description of Results

The evidence rejects a simple main-effect interpretation of platform membership. Platform membership is positive in market-FE models, but the same coefficient becomes small and insignificant in unit-FE models. This pattern suggests that platform membership is not best understood as a uniform utilisation booster. Instead, the stronger empirical regularities appear in interaction terms between platform membership and marketing-mix variables.

The most consistent baseline result is the pricing-friction effect. Unlock fees become more detrimental in platform contexts, especially once the fee crosses salient high-fee thresholds around EUR 1.00 and EUR 1.20. The threshold results are strongest for Local MaaS and multi-platform contexts. At the EUR 1.00 cutoff, Local MaaS has a market-FE coefficient of -0.229 and a unit-FE coefficient of -0.049. Multi-platform membership has a market-FE coefficient of -0.146 and a unit-FE coefficient of -0.099. Both effects are statistically significant. Similar estimates appear at EUR 1.20.

The leave-one-city tests show that these high-fee platform-type results are not driven by a single city. For Local MaaS, all 13 leave-one-platform-city tests remain negative and significant for EUR 0.99, EUR 1.00, and EUR 1.20. For multi-platform membership, the EUR 1.00 and EUR 1.20 thresholds also remain negative and significant in all 13 city-drop models. The effect is therefore not only a Berlin, Hamburg, or Munich artifact.

Outcome robustness further strengthens the Local MaaS pricing-friction result. The Local MaaS high-fee penalty appears not only in capacity utilisation but also in trip volume. For log trips, the EUR 1.00 Local MaaS effect is -1.014 in market FE and -0.130 in unit FE. Multi-platform evidence is cleaner for capacity utilisation than for trip volume, because the log-trip unit-FE coefficient is not significant.

The extended feature models sharpen the interpretation. The relative unlock-fee gap to the cheapest same-mode competitor is the strongest new feature. In Local MaaS contexts, the interaction is -0.089 in market FE and -0.030 in unit FE. In multi-platform contexts, it is -0.091 in market FE and -0.041 in unit FE. These estimates indicate that platform contexts are especially punitive when a provider has a higher upfront access fee than direct same-mode alternatives.

Other features are weaker or more bounded. Same-mode competition is negative for Local MaaS in market FE and marginal in unit FE, but it is less robust than relative unlock-fee position. Per-minute price gaps are also weaker than unlock-fee gaps. Active-vehicle share helps in some platform contexts, especially within unit, but it does not eliminate high-fee penalties. Night-oriented usage contexts are negatively associated with Local MaaS platform effectiveness. Promotion results are promising but locally bounded: subscriber-free-minutes perform well in multi-platform models but are based on one treated city-operator cell, Nürnberg-VOI.

6 Interpretation and Analysis

The results support a comparison-salience theory of platform membership. Platforms do not simply create demand by adding another channel. Instead, they change the information environment in which users evaluate mobility options. When multiple providers or modes are visible in an integrated interface, upfront price differences become more salient. This is why the unlock-fee effects are stronger and more consistent than per-minute price effects. Unlock fees are immediate, discrete, and easy to compare before the trip begins. Per-minute prices are less salient because their total cost implications are uncertain until trip duration is known.

Local MaaS and multi-platform memberships are the key contexts. Large aggregator-only membership does not show the same stable high-fee penalty. This matters theoretically: the relevant mechanism is not simply commercial aggregator exposure. Rather, the strongest effects occur when shared mobility is embedded in a broader urban mobility choice set. Local MaaS interfaces and multi-platform exposure plausibly make outside options more visible and comparable. In that setting, a high or relatively high unlock fee becomes a stronger deterrent.

The new relative-price features substantially improve the story. Absolute high-fee thresholds show that salient fees matter. Relative same-mode unlock-fee gaps show why they matter in a platform context: users do not evaluate the fee in isolation; they can compare it against direct alternatives. This provides a more marketing-specific account of platform membership. The platform does not merely moderate price sensitivity; it changes the competitive frame through which price is interpreted.

The vehicle-type breadth result adds a useful product-side counterpoint. Broader product assortment appears to buffer Local MaaS high-fee penalties. This suggests that marketing mix elements can interact: pricing friction is harmful, but product breadth can partially offset it by improving modal fit. The finding should not be overstated as a universal product-breadth effect, but it gives the paper a more balanced story than “platforms make fees bad.” Platforms amplify pricing frictions, yet product design can soften the effect.

Promotion is theoretically aligned with the mechanism but empirically less clean. The subscriber-free-minutes result suggests that usage-cost-reducing promotions can work well in multi-platform contexts. However, the evidence is localized to Nürnberg–VOI. Therefore, promotion should be used as mechanism evidence rather than as a central causal claim. Future data with campaign start and end dates across multiple cities would allow a stronger test.

The competition findings are also better interpreted as boundary conditions than as the main effect. Same-mode competition matters moderately in Local MaaS contexts, but the relative unlock-fee gap is stronger. This implies that competition per se is not enough. What matters is whether competition makes a specific marketing-mix disadvantage visible. In other words, competition enables comparison, but the marketing-mix attribute being compared remains central.

7 Implications for the Paper

The strongest paper-facing claim is:

Platform membership reshapes marketing-mix effectiveness by increasing the salience of comparable mobility alternatives. In Local MaaS and multi-platform contexts, providers with high or relatively high unlock fees experience lower capacity utilisation. This effect is not a generic platform main effect, not a Free Now-only result, and not driven by a single city.

The main empirical table should therefore prioritize: (1) high unlock-fee thresholds for Local MaaS and multi-platform membership; (2) relative same-mode unlock-fee gaps; (3) outcome robustness for trip volume; and (4) city-drop robustness. Product breadth can be used as a secondary moderation result. Promotion and provider-level pass features should be framed as mechanism evidence and future data opportunities unless richer city-date coding is added.

8 Limitations

Several limitations should be explicit. First, the data are observational and city–operator–day based. The results identify robust conditional associations rather than randomized treatment effects. Second, some platform types are structurally tied to cities and providers, making unit-FE evidence especially important. Third, provider-level external features such as pass availability, reservation, and parking/geofence rules lack enough within-unit time variation for robust estimation. Fourth, promotion results are based on few treated campaign contexts. Finally, same-platform competitor count is currently too collinear with platform membership and city/operator structure to be interpreted as a clean moderator.